

Can MSI Worksheets be Enhanced?

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Purpose

The goal was to develop and evaluate a web-based prototype of an interactive MSI Worksheet. The name used to describe the interactive Worksheet is “ErgoCheck”.

Context

MSI claims

In B.C. the MSI claim count and claim percentages have remained almost at the same level for 10 years – between 31% and 34% (Figure 1).

Knowing that MSI claims have been at the same level for over 10 years is a reason for deciding that some additional user-friendly and useful MSI tools are needed.

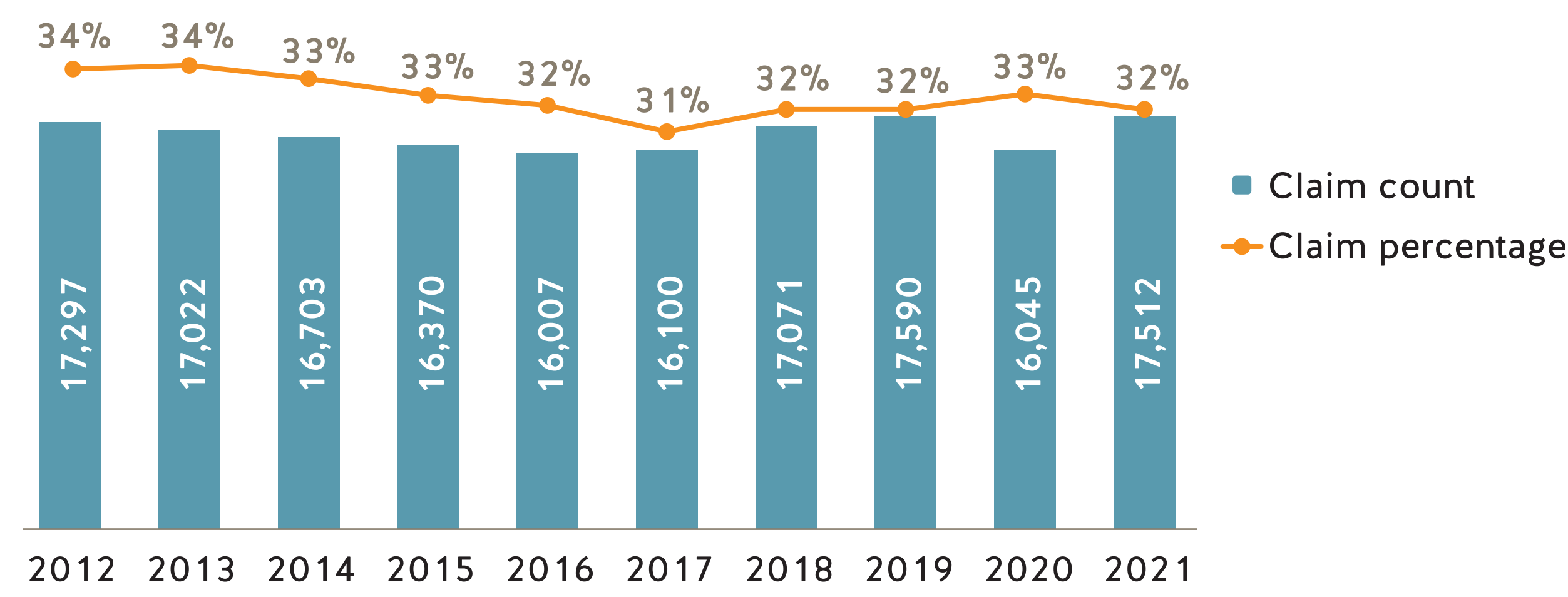


Figure 1: The MSI claim count and the percentage of claims in B.C. from 2012 to 2021.

Development of ErgoCheck

The initial version of ErgoCheck was developed by Liam Smith and Geoff Wright while both were employed at the City of Vancouver.

In September 2021, Geoff Wright became an employee at WorkSafeBC. On February 7, 2022, a legal document was signed by the City of Vancouver agreeing to the further development of ErgoCheck by WorkSafeBC.

Outcomes



Scan the QR code to access ErgoCheck

ErgoCheck

ErgoCheck asks the user a series of questions about situations where MSI hazards could be present in their workplace. The survey questions are comprised of 6 parts.

The priority ranking for each activity where there is an identified MSI risk factor is determined by a standard “Risk Assessment Matrix” (likelihood of the activity occurring and the level of discomfort/injury). The value assigned to the respondent’s answer is based on an ergonomist’s professional opinion on the likelihood of that activity being associated with the risk of musculoskeletal discomfort or injury and the corresponding level or severity of the injury corresponding to that activity.

ErgoCheck generates an MSI Risk Factor Report that identifies, categorizes, and scores the five primary MSI risk factors.

Relevant text is also populated in a risk identification table. For example, if a user answers “Frequently” to the question of “How much total time do you lean or rest on sharp objects or edges?” the risk identification table would populate “Frequent exposure with sharp edges/objects” in the “Action Needed” group.

There is also an Overall Musculoskeletal Risk (OMR) score calculated for each of the five factors with associated “Risk Priority Statements”.

Development of ErgoCheck

The initial version of ErgoCheck required a two-step process since the answers provided had to be manually transferred to the Excel file for analysis and generation of the reports.

The beta version of ErgoCheck was revised to include key user feedback and to incorporate the WorkSafeBC brand. Questions were reworded to be clearer and more accurate.

WorkSafeBC’s Experience Design department conducted a second phase of focus group testing involving prevention officers and employers. Additional improvements to the beta version of ErgoCheck were added.

As was the case with the other focus groups, the number of employers, supervisors, and workers who participated in the testing of the latest beta version of ErgoCheck and who submitted the follow-up survey was small.

Discussion

The development of ErgoCheck

The purpose of creating the alpha version of ErgoCheck was to demonstrate how the process for identifying common hazards and assessing the risks associated with musculoskeletal injuries could be automated. Both authors were aware of the limitations inherent in using Excel, but this initiative was considered the first step in demonstrating how a web-based tool that would assist employers and workers could be developed.

Focus group comments about ErgoCheck

The focus groups provided many suggestions for enhancements. Their suggestions and recommendations provide valuable insights and considerations in the development of a musculoskeletal injury prevention tool and deserve additional comment and discussion.

a) Industry-specific information

One focus group recommended more generic images and industry-specific versions based on the user-specified role or the identified industry. Another focus group also expressed interest for industry-specific versions. The suggestion of industry-specific enhancements is reasonable.

Implementation of this request goes beyond the current scope of the ErgoCheck initiative since it requires the commitment of additional resources. It would be appropriate to implement industry-specific content as enhancements once a basic tool has been released.

b) Actionable measures and tips

Once a basic web-based version was developed, discussions with the platform developer started regarding how WorkSafeBC staff could add appropriate links easily. Enabling WorkSafeBC to add links was considered the best plan since it would allow adding and maintaining the links to its library of educational and instructional materials and tools.

For example, MSI specific publications and MSI tools are available at worksafebc.com. Find publications by searching ‘Ergonomics’. Find MSI tools such as [Lift/Lower Calculator](#) and [Push/Pull/Carry Calculator](#).

c) Focused on one task and one person

The participants taking part in the beta Concept Test did not receive any instruction or information about what ErgoCheck does — or what it can do. The focus-group comments show the need to develop an informational booklet on the ways ErgoCheck can be used.

While ErgoCheck can process data provided for a general job category, it is structured to identify, evaluate, and compare the MSI risk factors associated with a duty, task, or work activity.

The concern about only reflecting one worker’s perspective can be resolved by asking other workers who perform the same assigned work task to complete the ErgoCheck survey. Combining the individual reports would highlight the similarity in the risk factors present when performing the work.

d) Language

Producing the ErgoCheck survey questions in the user’s preferred language is easier than translating the generated report. The important (and cost-saving) point to note is

that it may only be necessary to translate the questions. Once a question in the user’s preferred language is answered, their response is captured and the report generated. Although it is not an ideal situation, an individual user should be able to locate an individual who can translate the written report for them.

e) Assessment by an expert

Most employers and workers would prefer an assessment by an expert. However, WorkSafeBC does not have the resources or the mandate to provide this level of support to employers and workers.

After the latest version of ErgoCheck had been used, a follow-up assessment was conducted. It was found that the information provided helped determine which issues to focus on. The ErgoCheck report enhanced the level of communication between the employer, the workers, the WorkSafeBC occupational health and safety consultant and the WorkSafeBC ergonomist.

ErgoCheck was developed for use by workers, supervisors, and employers. While it was not developed for WorkSafeBC officers to use directly, the officers would likely be interested in seeing and discussing the results of a report generated by a worker or employer.

f) Reluctance to share results

It is interesting that the survey respondents were more likely to share the results with their co-workers rather than their supervisor or their Joint Health and Safety Committee. This outcome is more of a comment about the culture of safety within the organization than a limitation of ErgoCheck.

A Participatory Ergonomics Process plays a key role in achieving effective ergonomics interventions (Wright, 2017). It is important to involve senior leadership, joint committees, and workers in the process (WorkSafeBC, 2023). The important components for this process are management commitment and worker participation.

ErgoCheck was developed to support and enhance the communication about MSI risk between supervisors and workers and support the discussions on the need for interventions to control MSIs between a supervisor and employer.

Case study to practice summary

The advantage of an interactive MSI Worksheet is that it immediately engages employers, supervisors, and workers. It provides instant and objective findings on MSI risks, and supports fact-based conversations between safety officers, employers, and workers. It also increases awareness and education on MSIs, especially for young and new workers. Once developed beyond its current identification and assessment stage, ErgoCheck would link to relevant WorkSafeBC publications.

While ErgoCheck is currently more tailored towards workers who engage in manual labour, it could eventually be used to assess worker-specific industries such as construction, manufacturing, forestry, healthcare, and roles within an office environment.

Acknowledgement

A necessary and important piece in the development of ErgoCheck is the overall support provided by management at WorkSafeBC and the availability, enthusiasm, and commitment of WorkSafeBC’s professional staff. The existence of these behind-the-scenes resources and the willingness of individuals, workers, and employers to participate in focus group and testing sessions and the platform developer (maru/matchbox®) were essential pieces in the development of ErgoCheck into its current form.

References

WorkSafeBC (2023). Enhancing Safety Culture. [Enhancing safety culture \(sharepoint.com\)](#)

Wright, G. R. (2017). A Practitioner’s Perspective on adding an SMS approach to the Participatory Ergonomics Process. Presented at the 48th Annual ACE Conference & 12th International Symposium on Human Factors in Organizational Design and Management. Banff Centre for Arts and Creativity, Banff, Alberta July 31 to August 3, 2017.